Week		Content	Reading	HW Assigned	PA Assigned
week1	Tuesday, September 1, 2026	Course policies, introduction to Th. Of Comp. First look at Finite State Automata (FSA)	Chapter 0		
	Thursday, September 3, 2026	Terminology and formal defnition of FSAs state diagrams, regular languages (RL) operations on RLs	Chapter 1.1	PS1 Out Due Thu, 09/17, 11:59pm	
week 2	Tuesday, September 8, 2026	Closure of RLs under operations of union and intersection, the product machine	Chapter 1.1		PA1 Out Wed, 10/07, 11:59pm
	Thursday, September 10, 2026	Non-deterministic Finite Automata, closure under concatenation & Kleene star	Chapter 1.2		
week 3	Tuesday, September 15, 2026	Other regular operations Equivalence of NFAs and DFAs, Subset Construction	Chapter 1.2		
	Thursday, September 17, 2026	NFAs whose DFAs have exponential size, Regular Expressions	Hopcroft, et al. 2.3.6 Chapter 1.3	PS2 Out Due Thu, 10/01, 11:59pm	
week 4	Tuesday, September 22, 2026	Conversion from REs to NFAs/DFAs GNFAs	Chapter 1.3		
	Thursday, September 24, 2026	Conversion from DFAs → REs The State Elimination Algorightm for GNFAs	Chapter 1.3		
week 5	Tuesday, September 29, 2026	Distinguishable & equivalent states in a DFA, Distinguishability Lemma, the Quotient automaton	Hopcroft, et al. 4.4		
	Thursday, October 1, 2026	State minimization algorithm for DFAs, the Pumping Lemma for regular languages	Hopcroft, et al. 4.4.3 Sipser 1.4	PS3 Out Due Thu, 10/15, 11:59pm	
week 6	Tuesday, October 6, 2026	Examples of non-regular languages, using the the Pumping Lemma to prove non-regularity	Chapter 1.4		PA2 Out Due Thu, 11/05, 11:59pm
	Thursday, October 8, 2026	Context Free Grammars, Leftmost Derivations Parase Trees	Chapter 2.1		
week 7	Tuesday, October 13, 2026	CFG Ambiguity, CFL Closure Properties, Chomsky's normal form,	Chapter 2.1		
	Thursday, October 15, 2026	Review		PS4 Out Due Thu, 10/29, 11:59pm	
week 8	Tuesday, October 20, 2026	MIDTERM EXAM			
	Thursday, October 22, 2026	Nondeterministic Pushdown Automata & CFLs	Chapter 2.2		
week 9	Tuesday, October 27, 2026	Equivalence of NPDAs and CFGs	Chapter 2.2		
	Thursday, October 29, 2026	The Pumping Lemma for CFLs Examples of non-CFL languages	Chapter 2.3	PS5 Out Due Thu, 11/12, 11:59pm	
week 10	Tuesday, November 3, 2026	Turing Machines	Chapter 3.1		PA3 Out Due Fri, 12/04, 11:59pm
	Thursday, November 5, 2026	Multi-tape Machines & Their 1-Tape Simulators	Chapter 3.2		
week 11	Tuesday, November 10, 2026	Nondeterministic Turing Machines and Their Deterministic Simulators	Chapter 3.2		
	Thursday, November 12, 2026	The Universal Machine, Decidable Languages	Chapter 4.1	PS6 Out Due Tue, 11/24, 11:59pm	
week 12	Tuesday, November 17, 2026	Undecidable & Unrecognizable Languages	Chapter 4.2		
	Thursday, November 19, 2026	Additional Undecidable Problems Complexity of Turing Machines	Chapter 4.2, 7.1		
week 13	Tuesday, November 24, 2026	Classes P and NP	Chapter 7.2, 7.3 (definition of class NP is from recommended texbook, page 431, Section 10.1.3)	PS7 Out Due Mon, 12/07, 11:59pm	
	Thursday, November 26, 2026	Thanksgiving!			
week 14	Tuesday, December 1, 2026	Polynomial Reductions	Chapter 7.3, 7.4		PA3 Due Due Fri, 12/04, 11:59pm
	Thursday, December 3, 2026	Cook's Theorem, NP-Complete Problems	Chapter 7.4		
week 15	Tuesday, December 8, 2026	Review			
	Thursday, December 10, 2026	FINAL EXAM		PS7 Due Due Mon, 12/07, 11:59pm	